

CampusEats — Service Design

CS543 Web Services — Assignment 2: Services, Contracts & Schema

Task 1: System Capabilities

1. **Discover Food** — Allows students to discover available campus food services.
2. **View Food Service** — Allows students to view information about campus food services.
3. **Place Order** — Allows students to place food orders.
4. **Process Order** — Allows food services to process submitted orders.
5. **Manage Orders** — Allows the system and food services to manage food orders.
6. **Manage Food Services** — Allows campus food services to be managed by the system.

Task 2: Service Design

CampusEats is divided into five services based on data ownership:

Service	Owned data
Identity Service	Student, Vendor, and Administrator data.
Catalog Service	Restaurant, Food, and Menu Item data.
Order Service	Order and Order Item data.
Payment Service	Payment data.
Delivery Service	Delivery data.

Task 3: Service Contracts

Identity Service

Operation	Input	Output	Errors
register()	user details	registration result	invalid details; user already exists
login()	login credentials	authentication result	invalid credentials
manage()	user management request	updated user information	unauthorized request; invalid request

Catalog Service

Operation	Input	Output	Errors
browse()	search or browse criteria	available food services	invalid criteria; service unavailable
view()	food item or service request	food/service details	item not found
select()	selected food items	selected item information	item unavailable; invalid selection
manage()	catalog management request	updated catalog information	unauthorized request; invalid request

Order Service

Operation	Input	Output	Errors
order()	selected food and order details	order confirmation	invalid order; unavailable item
cancel()	order cancellation request	cancellation result	order not found; order cannot be cancelled
manage()	order management request	updated order status	unauthorized request; invalid request

Payment Service

Operation	Input	Output	Errors
pay()	payment request	payment result	payment failed; invalid payment request

Delivery Service

Operation	Input	Output	Errors
deliver()	delivery request	delivery status	delivery unavailable; invalid request
track()	tracking request	current delivery status	delivery not found

Task 4: placeOrder Specification

Operation: placeOrder

The **placeOrder** operation allows a student to submit an order for selected food items.

Inputs

- Student identity
- Selected food items
- Quantities
- Order details
- Payment details
- Delivery details

Outputs

- Order confirmation
- Order identifier
- Order status
- Payment status
- Delivery status

Error Cases

- Student is not authenticated
- Selected food item is unavailable
- Invalid quantity
- Invalid order details
- Payment fails
- Delivery service is unavailable
- Order cannot be created

Internal Details Hidden from Callers

1. Validating the student and order details.
2. Checking the availability of selected food items.
3. Creating the order.
4. Processing the payment.
5. Preparing the order for delivery.
6. Creating or updating delivery information.
7. Updating the order status.
8. Returning the final order result to the caller.

Task 6: Service Validation

Each CampusEats service was checked against the five required service properties: reachable, self-contained, contract-based, independent, and loosely coupled.

Service	Reachable	Self-contained	Has Contract	Independent	Loosely Coupled
Identity Service	Yes	Yes	Yes	Yes	Yes

Service	Reachable	Self-contained	Has Contract	Independent	Loosely Coupled
Catalog Service	Yes	Yes	Yes	Yes	Yes
Order Service	Yes	Yes	Yes	Yes	Yes
Payment Service	Yes	Yes	Yes	Yes	Yes
Delivery Service	Yes	Yes	Yes	Yes	Yes

Validation Notes

- Identity Service: Provides access to student, vendor, and administrator data through its own service boundary.
- Catalog Service: Manages restaurant, food, and menu item data independently.
- Order Service: Manages orders and order items and exposes operations through a defined contract.
- Payment Service: Owns payment data and handles payment-related operations separately from other services.
- Delivery Service: Owns delivery data and manages delivery status independently.

All five services satisfy the required properties. The services communicate through defined contracts rather than directly sharing their internal implementation or database tables.

Note: The current design.md does not specify individual implementation members, so no member names have been invented in this PDF.